Corner cuts and monomial prefixes: sharp counts and prime realizations

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Abstract

Let $C_d(n)$ count the n-point subsets of the nonnegative integer lattice that can be strictly separated from their complements by an affine hyperplane. For every fixed $d \geq 2$, we prove

$$C_d(n) = \Theta_d\big(n^{d-1}(\log n)^{d-1}\big).$$

The same order holds for the number of complete ordered prefixes of length n, equivalently the strictly admissible finite monomial orderings. The set-count upper bound is due to Wagner. For the lower bound, a finite sieve produces primitive lattice simplices in separated axis boxes. Their primitive normal directions are distinguished at a prescribed cardinality by two integer patches on opposite sides of a boundary level. Summing over disjoint aspect ranges supplies the logarithmic factor. This gives a matching lower bound in every fixed dimension, including the three-dimensional order $n^2(\log n)^2$. An active-face dyadic decomposition gives the ordered-prefix upper bound. We also prove an optimal worst-case prime-realization bound: every ordered prefix is realizable over distinct primes above B with $\log \max_i p_i = O_d(n \log \max(B, 2))$, and some prefixes require this order. The latter argument combines a linear integer-score certificate with an established short-interval prime theorem. Qualitative prime realization and the finite-weight representation are prior results.

1 The counting problem and its scope

A nonempty finite set $D \subset \mathbb{Z}^d_{\geq 0}$ is a *corner cut* if

$$D = \{x \in \mathbb{Z}^d_{\geq 0} : w \cdot x < t\} \quad \text{for some } w \in \mathbb{R}^d_{>0},\ t > 0,$$

with no lattice point on the cutting hyperplane. Write $C_d(n)$ for the number of such sets of cardinality n, and put

$$A_d(n) = \sum_{j=1}^{n} C_d(j).$$

The parameter $C_d(n)$ counts sets, without their internal orders or representing weights. Every cut is an order ideal for the coordinate order. In particular, every coordinate of an element of an n-point cut is at most $n-1$, so these counts are finite.

These objects occur as the exponent sets of initial segments of smooth numbers. For distinct primes $p_1, \ldots, p_d$, comparing the products $\prod_i p_i^{x_i}$ is equivalent to comparing $\sum_i x_i \log p_i$. An exponent prefix is consequently a corner cut. Conversely, every corner cut has such a realization, even with all primes above any prescribed bound [11, Proposition 6.2]. Theorem 9 bounds the required prime sizes. Thus the full geometric class is exactly the class of

exponent-prefix sets as the ordered prime bases vary. This does not give a prime-distribution estimate or a faster random-access algorithm for a fixed prime set.

Corner cuts were introduced in connection with state polyhedra and Gröbner bases by Onn and Sturmfels [1]. The planar order $C_2(n) = \Theta(n \log n)$ is due to Corteel, Rémond, Schaeffer and Thomas [2]. For fixed d, Wagner [3] proved

$$\Omega_d(n^{d-1} \log n) \le C_d(n) \le O_d(n^{d-1}(\log n)^{d-1}). \tag{1}$$

His thesis [4, Chapter 3] gives the lower and upper arguments and suggests seeking greater completion multiplicity. Wagner also credits Rémond with independently obtaining the bounds in (1). We prove a matching lower bound, improving the lower logarithmic exponent by $d-2$ when $d \ge 3$. We count primitive normals in each of many aspect ranges and prove that they give distinct prefixes at every sufficiently large prescribed cardinality. The latter step, rather than a lift from a lower dimension, is the key to retaining every logarithmic factor.

Theorem 1. *For each fixed integer $d \ge 2$,*

$$C_d(n) = \Theta_d\big(n^{d-1}(\log n)^{d-1}\big).$$

Here is one explicit lower bound. Put

$$K_d = 4^{d+1}, \qquad T_d = [64d^2(K_d+1)]^d, \qquad R_0(d) = 4d^2(d+1),$$

$$n_*(d) = T_d 4^{R_0(d)}, \qquad \rho_d = 8^{-(d-1)}(100d)^{-d^2},$$

and

$$\lambda_d = \frac{\rho_d}{2^d(4T_d)^{d-1}[4(d+1)]^{d-1}(d-1)!}.$$

For every integer $n \ge n_*(d)$,

$$C_d(n) \ge \lambda_d n^{d-1}(\log_4 n)^{d-1}. \tag{2}$$

Summing gives $A_d(n) = \Theta_d(n^d(\log n)^{d-1})$. The pointwise conclusion matches (1) and strengthens its lower logarithmic exponent when $d \ge 3$. For $d = 2$, it recovers the known planar order.

We also count complete ordered prefixes, denoting their number by $O_d(n)$. These are precisely Vasiliev–Pavlov's strictly admissible finite monomial orderings [12, Definition 5]; the class and its enumeration are established background. Theorem 7 proves $O_d(n) = \Theta_d(n^{d-1}(\log n)^{d-1})$. Thus retaining the full internal order changes the total count by only a bounded factor in fixed dimension. Theorem 9 then gives a sharp worst-case arithmetic size bound for realizing any such word over prime bases. All constants may depend on d; the dimension is fixed throughout.

2 Primitive simplices in axis boxes

Fix an ambient integer $d \ge 2$ and set $\delta = 1/(100d)$. For integer side lengths $L_1, \dots, L_d$, put

$$m = \min_i L_i, \quad M = \max_i L_i, \quad Q = \prod_{i=1}^{d} L_i.$$

Define $B_i \subset \mathbb{Z}^d_{\ge 0}$ by taking the jth coordinate interval to be

$$[L_j, L_j + \lfloor \delta L_j \rfloor] \quad (j = i), \qquad [0, \lfloor \delta L_j \rfloor] \quad (j \ne i).$$

A d-by-$(d-1)$ integer matrix is called primitive if the greatest common divisor of its maximal minors is one.

Lemma 2. *Assume $M \le m^2$ and $m \ge 2^{8d^2}$. At least a fraction $8^{-(d-1)}$ of the ordered choices $p_i \in B_i$ have primitive difference matrix*

$$V = [p_2 - p_1 \ \cdots \ p_d - p_1].$$

The minor criterion and its equivalence with full column rank modulo every prime are standard [8, Propositions 3–4]. The point requiring proof here is uniformity for these moving, possibly anisotropic boxes.

Proof. Keep the same δ throughout a dimension induction. Let E_r be the event that the first r points, projected onto the first r coordinates, have primitive difference matrix. The event E_1 is automatic, with empty minor equal to one. At step $2 \le r \le d$, fix a primitive configuration of the previous $r-1$ projected points. Its difference matrix U has size $(r-1)$-by-$(r-2)$. Its vector a_0 of signed maximal minors is primitive, and U has rank $r-2$ modulo every prime.

Let u be the first $r-1$ coordinates of $p_r - p_1$, and put $D = \det[U \ u]$. This determinant is nonzero for every allowed choice. Indeed, the normalized row matrix of the previous $r-1$ projected points differs from the identity by at most $(r-1)\delta$ in maximum row-sum norm. The normal of their affine span, normalized to height one, has scaled coordinates within $(r-1)\delta/(1-(r-1)\delta)$ of one and is positive. The new projected point has every scaled coordinate at most δ; it is strictly on the origin side of that affine span. Also,

$$|D| \le (r-1)!\, 2^{r-1} M^{r-1} \le 2^{d^2} m^{2(d-1)}. \tag{3}$$

Every coordinate interval contains more than δm integers. For a prime ℓ, the equation $D \equiv 0 \pmod{\ell}$ is a nontrivial linear equation in the new projected point: at least one coordinate of a_0 is nonzero modulo ℓ. Its probability is at most $1/\ell + a/m$, where $a = 100d$.

Suppose $\ell \mid D$. Since U has full column rank modulo ℓ, there is a unique vector λ with $u \equiv U\lambda \pmod{\ell}$. Write z for the rth row of the old columns and u_r for the rth coordinate of the new column. The full r-by-$(r-1)$ matrix loses rank modulo ℓ precisely if

$$u_r \equiv z\lambda \pmod{\ell}. \tag{4}$$

Conditional on everything except the last coordinate of p_r, this is one residue class, of probability at most $1/\ell + a/m$. Thus the probability of failure at any particular prime is at most $(1/\ell + a/m)^2$.

Put $s = \lfloor (\log_2 m)/2 \rfloor$, $P = 2^s$. The small-prime union bound gives

$$\begin{aligned} \Pr(\text{failure at a prime } \ell \le P) &\le \sum_{\substack{\ell \le P \\ \ell \text{ prime}}} (1/\ell + a/m)^2 \\ &\le \frac{3}{4} + \frac{2a(s+1)}{2^{2s}} + \frac{a^2}{2^{3s}}. \end{aligned} \tag{5}$$

Here $\sum_{j=2}^{\infty} j^{-2} < 3/4$ follows by comparison with $1/4 + \int_2^{\infty} x^{-2}\,dx$. The cross term uses $H_P \le 1 + \log P \le 1 + s$, and the constant term uses at most P primes.

For large primes, first fix the new projected coordinates. Every bad prime $\ell > P$ divides the same nonzero integer D. By (3), their number is at most

$$\frac{\log_2 |D|}{s} \le 4(d-1) + \frac{d^2 + 4(d-1)}{s} \le b := 4d - 3 \quad \text{if } s \ge d^2 + 4(d-1).$$

For each of these finitely many primes, (4) bounds the conditional last-coordinate failure probability by $2^{-s} + a2^{-2s}$. Their total contribution is therefore at most $b(2^{-s} + a2^{-2s})$.

Define

$$E_d(s) = \frac{2a(s+1)}{2^{2s}} + \frac{a^2}{2^{3s}} + b(2^{-s} + a2^{-2s}). \tag{6}$$

This decreases for $s \geq 1$. For every integer $d \geq 2$, the choice $S_d = 4d^2$ satisfies $S_d \geq d^2 + 4(d-1)$ and $E_d(S_d) < 2^{-10} < 1/8$. For an explicit verification valid in every dimension, use $a \leq 2^{4d}$, $b \leq 2^{2d}$ and $4d^2 + 1 \leq 2^{3d}$. They give

$$E_d(4d^2) \leq 2^{1+7d-8d^2} + 2^{8d-12d^2} + 2^{2d-4d^2} + 2^{6d-8d^2} < 2^{-10}.$$

The four exponents decrease for $d \geq 2$, and at $d = 2$ are $-17, -32, -12, -20$.

Consequently the conditional probability of a nonprimitive extension is at most $3/4 + E_d(s) \leq 7/8$. To justify the conditioning across steps, the events $E_1, \ldots, E_{r-1}$ depend only on previous projected coordinates. After these are fixed, the old rth coordinates and the new point retain their product distributions. Thus

$$\Pr(E_r \mid E_1, \ldots, E_{r-1}) \geq 1/8.$$

Multiplication proves that $\Pr(E_1 \cap \cdots \cap E_d) \geq 8^{-(d-1)}$. This intersection implies E_d, as required. All bounds on large primes were finite before any limit was taken. □

3 Primitive normals at a prescribed size

For each primitive frame from Lemma 2, orient its cofactor normal a positively and put $H_a = a \cdot p_1$. Let P have rows p_i. Its column-scaled matrix differs from the identity by a matrix of maximum row-sum norm at most $d\delta = 1/100$. The determinant is positive and every eigenvalue lies within $1/100$ of one. Consequently

$$H_a = \det P, \qquad 2^{-d}Q \leq H_a \leq 2^d Q. \tag{7}$$

The solution of $p_i \cdot w = 1$ satisfies $|L_i w_i - 1| \leq d\delta/(1 - d\delta) < 1/10$. Since $w = a/H_a$, this gives

$$\frac{9}{10} < \frac{a_i L_i}{H_a} < \frac{11}{10}. \tag{8}$$

Lemma 3. *Under the hypotheses of Lemma 2, these frames give at least $2^{-d}\rho_d Q^{d-1}$ distinct primitive positive normals.*

Proof. Primitivity gives an integer z with $a \cdot z = 1$. The matrix $[V\ z]$ is unimodular, so the columns of V form a basis of the entire integer kernel of a. If two points of the same B_i lie in the selected affine plane, their difference is Vc, with $c \in \mathbb{Z}^{d-1}$. The submatrix in rows $2, \ldots, d$, after row scaling by the L_i, differs from the identity by at most $(d-1)\delta$ in maximum row-sum norm. Thus

$$\|c\|_\infty \leq \frac{\delta}{1 - (d-1)\delta} < 1,$$

so the two points agree. Each plane determines its selected simplex. Each box has at least $\delta^d Q$ points, so the number of distinct planes is at least $\rho_d Q^d$. For any fixed primitive normal, the integer height in (7) has at most $2^d Q$ possibilities. Division proves the claim. □

When one normal occurs at several heights, fix one witnessing simplex for it, and use that simplex to define H_a. The prefix below depends only on a, not on this choice of witness.

For a positive integer normal a, order lattice points by $(a \cdot x, x_1, \ldots, x_d)$, and let $D_n(a)$ be the first n. Let h be the least integer for which

$$F_a(h) := |\{x \in \mathbb{Z}^d_{\geq 0} : a \cdot x \leq h\}| \geq n.$$

The set $D_n(a)$ is a corner cut, including when the score-h level is only partly included. Indeed $h \le (n-1)\min_i a_i$, by considering a minimum-weight axis. Every point at score at most h lies in $\{0,\dots,n-1\}^d$. For $n>1$, the integer vector $u_i = n^{d-i}$ represents lexicographic order in that box, and $u\cdot x \le n^d - 1$ there. Taking $w = n^d a + u$ preserves all these levels and strictly orders the boundary level. Positivity keeps every higher a-level above them. A threshold between consecutive integer scores gives a strict representation. The case $n=1$ is immediate. We will use the weak separation property

$$b\cdot x \le b\cdot y \quad (x\in D_n(b),\ y\notin D_n(b)). \tag{9}$$

Lemma 4 (Fixed-size injectivity)**.** *Let a,b be primitive positive normals furnished by the same aspect box. Put $K = 4^{d+1}$. If the boundary height h of $D_n(a)$ satisfies*

$$h/H_a \ge 16d^2(K+1), \tag{10}$$

then $D_n(a) = D_n(b)$ implies $a = b$.

Proof. Choose the selected frame $V = [v_1\ \cdots\ v_{d-1}]$ for a. Since V is a basis of its integer kernel, an integer solution of $a\cdot z = 1$ can be reduced modulo V to give

$$z = p_1/H_a + Vc, \qquad |c_j| \le 1/2. \tag{11}$$

Each entry satisfies $|(v_j)_i| < 2L_i$, so $|z_i| \le (d+1)L_i \le 2dL_i$. Let $y_i = (h-1)/(da_i)$. Reduce $(h-1)z$ modulo V about y, obtaining an integer x_0 such that

$$a\cdot x_0 = h-1, \qquad x_0 - y = V\beta, \qquad |\beta_j| \le 1/2.$$

Both integer patches

$$\{x_0 + Vk : k\in\mathbb{Z}^{d-1},\ |k_j|\le K\}, \qquad \{x_0 + 2z + Vk : k\in\mathbb{Z}^{d-1},\ |k_j|\le K\} \tag{12}$$

are nonnegative. To see this coordinatewise, the displacement from y_i is less than $(2Kd + 5d)L_i$, while (8) and (10) give

$$y_i/L_i > \frac{h-1}{2dH_a} > 2Kd + 5d.$$

Their a-scores are respectively $h-1$ and $h+1$. Thus the first patch is entirely inside $D_n(a)$, the second entirely outside; the score-h tie is irrelevant.

Assume the two prefixes agree, and put $S = \sum_j |b\cdot v_j|$. By (9), the maximum b-score on the first patch cannot exceed the minimum on the second. Hence

$$KS \le b\cdot z \le b\cdot p_1/H_a + S/2. \tag{13}$$

For two normals in the same aspect family, (7) gives $H_b/H_a \le 4^d$, and (8) gives

$$\frac{b\cdot p_1}{H_a} < 4^d\frac{11}{10}(1+d\delta) < 2\cdot 4^d.$$

It follows that $(K-1/2)S < 2\cdot 4^d < K - 1/2$. But S is an integer, so $S = 0$. Thus b annihilates every column of V and is parallel to a. Primitivity and positivity force $a = b$. □

Lemma 5. *If $n/Q \ge T_d$ and $m \ge 2^{8d^2}$, then (10) holds. With $G = (d!n/Q)^{1/d}$, the maximum ith axis coordinate of $D_n(a)$ lies strictly between $GL_i/2$ and $2GL_i$.*

Proof. The disjoint half-open unit cubes based at the points counted by $F_a(t)$ contain the positive simplex of height t, and are contained in the simplex of height $t + \sum_i a_i$. Therefore

$$\frac{t^d}{d! \prod_i a_i} \le F_a(t) \le \frac{(t + \sum_i a_i)^d}{d! \prod_i a_i} \qquad (t \ge 0).$$

Use $F_a(h-1) < n \le F_a(h)$ and (8) to get

$$\frac{9}{10} G - \frac{11d}{10m} < \frac{h}{H_a} < \frac{11}{10} G + \frac{1}{H_a}.$$

Since $m \ge 10d$ and $G \ge 64d^2(K_d + 1)$, this implies $4G/5 < h/H_a < 6G/5$, and also (10). If M_i is the axis maximum, all integer axis points below score h are included and none above it are included. Thus $h/a_i - 2 < M_i \le h/a_i$. Combining these inequalities with (8) gives $GL_i/2 < M_i < 2GL_i$. □

4 Summing aspect ranges

Proof of Theorem 1. The upper bound is Wagner's theorem [3]; see also [4, Theorem 3.15]. For the lower bound, take R maximal with $Q = 4^R \le n/T_d$. For every tuple

$$r_1 + \cdots + r_d = R, \qquad r_i \ge \lceil R/(d+1) \rceil,$$

put $L_i = 4^{r_i}$. When $n \ge n_*(d)$, these tuples satisfy $R \ge R_0(d)$, $\min_i L_i \ge 2^{8d^2}$, and $\max_i L_i \le (\min_i L_i)^2$. Lemmas 3–5 therefore give at least $2^{-d} \rho_d Q^{d-1}$ different n-point cuts per tuple.

The value G in Lemma 5 is common to all tuples. Two distinct tuples differ by a factor of at least four in one L_i. The open observable ranges $(GL_i/2, 2GL_i)$ are disjoint for that coordinate, so no cut is counted twice across tuples.

Put $r = R - d\lceil R/(d+1) \rceil$. The number of tuples is

$$\binom{r + d - 1}{d - 1} \ge \frac{R^{d-1}}{[2(d+1)]^{d-1}(d-1)!},$$

since $r \ge R/(d+1) - d \ge R/[2(d+1)]$. Also $Q > n/(4T_d)$ and $R \ge (\log_4 n)/2$. For the latter explicit bound, $d^2 \le 4^{d-1}$ gives $T_d \le 4^{2d^2+4d}$, and $R_0(d) \ge 2d^2 + 4d + 2$. Multiplying the counts yields (2). □

5 Counting complete ordered prefixes

Let $\mathcal{O}_d(n)$ consist of ordered tuples $W = (x_1, \ldots, x_n)$ of the first n points of $\mathbb{Z}^d_{\ge 0}$ in increasing $w \cdot x$, as positive vectors w with rationally independent coordinates vary. Put $O_d(n) = |\mathcal{O}_d(n)|$. Coordinates are labeled. Forgetting the internal order maps this class onto the n-point corner cuts. Indeed, strict separation is controlled by the finitely many minimal excluded points, so a separator may be perturbed to have rationally independent coordinates.

We record explicitly the known connection with global monomial orders. The statement concerns each finite initial interval, and does not assert that one scalar vector represents an entire infinite monomial order.

Lemma 6 (Finite-weight representation)**.** *The words in $\mathcal{O}_d(n)$ are exactly the length-n initial intervals of global monomial orders.*

Proof. A positive rationally independent vector gives a total, multiplicative well-order on monomials, since only finitely many exponents have bounded score. Conversely, Robbiano's representation [13] expresses a global monomial order by lexicographic comparison of finitely many real linear forms $u_1, \dots, u_s$. For an initial interval W, put $D = \{x_1, \dots, x_n\}$ and let E be the coordinatewise minimal points outside D. The set E is finite: all its coordinates are at most one more than the corresponding coordinate maximum of D. Each vector in the finite family

$$\{e_i : 1 \le i \le d\} \ \cup \ \{x_{j+1} - x_j : 1 \le j < n\} \ \cup \ \{b - x_n : b \in E\}$$

has lexicographically positive image under these forms. Consequently $w(\varepsilon) = u_1 + \varepsilon u_2 + \cdots + \varepsilon^{s-1} u_s$ is positive on the whole family for sufficiently small $\varepsilon > 0$. The unit vectors make its coordinates positive; the other inequalities give the prescribed order and put the entire excluded border after it. Every excluded point dominates a border point, so positivity controls the infinite complement. A small generic perturbation preserves these finite inequalities and gives rational independence. □

Theorem 7 (Ordered-prefix count)**.** *For every fixed $d \ge 2$,*

$$C_d(n) \le O_d(n) = O_d\Big(n^{d-1}(\log(n+1))^{d-1}\Big).$$

In combination with Theorem 1, this gives $O_d(n) = \Theta_d(n^{d-1}(\log n)^{d-1})$.

Proof. The case $n = 1$ is immediate. For a witnessing weight w, put $t = w \cdot x_n > 0$, and restrict to the active coordinates $J = \{i : w_i \le t\}$. Write $r = |J| \ge 1$. All other coordinates vanish throughout the prefix. In this face put

$$b_i = t/w_i \ge 1, \qquad L_i = 2^{\lfloor \log_2 b_i \rfloor}, \qquad Q = \prod_i L_i = 2^R.$$

The simplex of height t is covered by the unit cubes based at its n lattice points, so $\prod_i b_i \le r!n$. The bounding box gives $n \le \prod_i (\lfloor b_i \rfloor + 1) \le 2^r \prod_i b_i$. Since $L_i \le b_i < 2L_i$,

$$n < 4^r Q, \qquad Q \le r!n. \tag{14}$$

For fixed r, the integer R occupies a bounded-width interval and is at most $\log_2 n + \log_2(r!)$. For each R, there are $\binom{R+r-1}{r-1}$ possible tuples of nonnegative integer exponents. Thus the number of aspect tuples is

$$O_r\big((\log(n+1))^{r-1}\big).$$

The lower product bound in (14) is essential here.

For one such tuple, all prefix points lie in $G_L = \prod_i \{0, \dots, 2L_i - 1\}$. Comparisons within this box change on central hyperplanes $w \cdot v = 0$, where $v \ne 0$ is an integer vector with $|v_i| \le 2L_i - 1$. Up to sign, there are at most

$$\frac{\prod_i (4L_i - 1) - 1}{2} \le 4^r Q \le 4^r r!n$$

such hyperplanes. An arrangement of $m \ge 1$ central hyperplanes in $\mathbb{R}^r$ has at most

$$2 \sum_{j=0}^{r-1} \binom{m-1}{j} = O_r(m^{r-1})$$

chambers. The bound follows by adding one hyperplane at a time and inducting on the restricted central arrangement; degeneracies only reduce the count.

Within a chamber the order on G_L is fixed, and any global prefix whose witness has this chart must be its first n points. Thus there is at most one word per chart and chamber. Charts that are not realized, and words counted in more than one chart, only overcount. There are consequently $O_r(n^{r-1}(\log(n+1))^{r-1})$ words for each active set J. Summing over its $\binom{d}{r}$ choices and over $1 \le r \le d$ proves the upper bound. The surjection onto corner cuts gives the lower bound $C_d(n) \le O_d(n)$. □

In particular, the mean number of admissible internal orders per n-point corner cut is $O_d(n)/C_d(n) = \Theta_d(1)$. This is an average statement and gives no constant bound for each shape. The upper argument also recovers Wagner's set-count upper order; its original attribution is unchanged. Wagner also bounds all face flags of the fixed-size corner-cut polyhedron [4, Section 3.5]. These are different from full ordered prefixes. For example, one planar cut of size 50 has five strict orders, realized by weights $(60, 49)$, $(84, 71)$, $(112, 97)$, $(144, 127)$ and $(18, 17)$. They give the same exposed vertex $(130, 155)$, which has only two incident maximal face flags. Thus identifying each word with a maximal flag at its final-cut vertex does not give the ordered bound. The chart proof above handles internal orders directly. This example makes no claim to exclude every other deduction from Wagner's flag theorem.

6 An optimal prime-size bound

Qualitative realization by distinct prime bases follows from Vietri's finite-inequality theorem [11, Proposition 6.2], including any prescribed lower bound on the primes. Consecutive comparisons and the minimal excluded border specialize that result to ordered prefixes. We now bound the worst-case size of the realizing primes.

Lemma 8 (Linear integer-score certificate)**.** *Put $c_d = d(d!)^2 2^{d-1}$. Every $W \in \mathcal{O}_d(n)$ has a positive integer weight vector A realizing exactly the same ordered prefix, with all included scores distinct. Its final score $S = A \cdot x_n$ satisfies*

$$n - 1 \le S \le c_d n, \qquad \max_i A_i \le S + 1. \tag{15}$$

For $n = 1$, take $A = (1, \dots, 1)$ and $S = 0$.

The proof uses a separation LP, determinant bounds and a simplex product estimate. These are established methods [1, 3]; we include the certificate bound as a supporting lemma, without a separate claim of a new general weight-compression method.

Proof. Assume $n > 1$, choose a strict positive witnessing vector w, and put $T = w \cdot x_n$. Restrict to the r active coordinates $w_i \le T$, and write $B_i = \lfloor T/w_i \rfloor \ge 1$. In this face $D = \{x_1, \dots, x_n\}$ is the set of all lattice points of score at most T. Its based unit cubes cover the simplex, and hence

$$\prod_i B_i \le \frac{T^r}{\prod_i w_i} \le r!n.$$

Let E be the minimal excluded border in this r-dimensional face. Each $b \in E$ has $b_i \le B_i + 1$, and every excluded point dominates an element of E. The finite system

$$z_i \ge 1, \quad (x_{j+1} - x_j) \cdot z \ge 1 \ (1 \le j < n), \quad (b - x_n) \cdot z \ge 1 \ (b \in E)$$

is feasible after scaling w. Minimize $\sum_i z_i$; bounded sublevel sets are compact, so the optimal face has a vertex. At that vertex choose r independent tight rows, giving an integer matrix

M with $Mz = \mathbf{1}$. Every entry in column i has absolute value at most $2B_i$. Cramer's rule gives positive integers $A_i = |\det M| z_i$ with

$$A_i \le r! 2^{r-1} \prod_{j \ne i} B_j.$$

All unit-margin constraints remain satisfied. Thus they give the prescribed order and strictly exclude the entire complement, and

$$S \le \sum_i B_i A_i \le r(r!)^2 2^{r-1} n \le c_d n.$$

Each active unit vector belongs to D, so $A_i \le S$. Restore inactive coordinates with weight $S+1$; every point using one of them is excluded. Finally, the n distinct nonnegative integer scores start at zero, so $S \ge n-1$. □

For $B > 0$ and $W \in \mathcal{O}_d(n)$, let $h_B(W)$ be the minimum of $\log \max_i p_i$ over ordered tuples of distinct primes $p_i > B$ realizing W. Let $H_d(n, B) = \max_{W \in \mathcal{O}_d(n)} h_B(W)$. All logarithms in this section are natural. Qualitative realization ensures these minima exist; there are finitely many words.

Theorem 9 (Sharp worst-case prime height)**.** *For each fixed $d \ge 2$, uniformly for $n \ge 2$ and $B > 0$,*

$$H_d(n, B) = \Theta_d\big(n \log \max(B, 2)\big).$$

More explicitly, put $\alpha = \max\{30, d, \log \max(B, 2) + 1\}$. Then

$$(n-1) \log \max(B, 2) \le H_d(n, B) < (\alpha + \log 2)(c_d n + 1) + 1. \tag{16}$$

The same bounds hold for the worst-case minimum when only an underlying n-point corner-cut set is prescribed.

Proof. Take the certificate of Lemma 8, and sort its coordinates temporarily as $m = A_1 \le A_2 \le \cdots \le A_d$. For distinct i, j,

$$\operatorname{lcm}(A_i, A_j) > S. \tag{17}$$

Otherwise two distinct pure-axis lattice points of that common score would both be included, contradicting their strict integer order. Consequently $A_j^2 \ge m A_j > n-1$ for every $j \ge 2$. Thus at most one weight can be as small as $\sqrt{n-1}$.

Our arithmetic input is Dusart's unconditional estimate [14, Proposition 5.4]: for real $x \ge 89693$ there is a prime $x < p \le x(1 + 1/(\log x)^3)$. In particular, for $x \ge e^{30}$,

$$x < p < x\left(1 + \frac{1}{25(\log x)^2}\right). \tag{18}$$

Put $\varepsilon = 1/[25\alpha^2(n-1)]$. Bertrand's postulate [6] gives a prime $e^{\alpha m} < p_1 < 2e^{\alpha m}$. Define

$$T = \frac{\log p_1}{m}, \quad \alpha < T < \alpha + \log 2, \qquad X_j = e^{TA_j}(1+\varepsilon)^{j-2} \quad (2 \le j \le d).$$

By (17), $\log X_j > \alpha\sqrt{n-1}$ and $X_j \ge e^{30}$. Formula (18) supplies

$$X_j < p_j < X_j(1+\varepsilon).$$

The intervals are disjoint since $A_{j+1} \ge A_j$, even if some larger weights agree; also $X_2 \ge p_1$. Hence all chosen primes are distinct and greater than B. Writing $\eta_i = \log p_i - TA_i$, we have

$$\eta_1 = 0, \qquad 0 < \eta_j < (d-1)\varepsilon \quad (j \ge 2).$$

Every point x of an n-point downset has $\sum_i x_i \le n-1$, by following a unit-step path from zero to x. It follows that

$$0 \le \eta \cdot x < \frac{d-1}{25\alpha^2} < \alpha < T \qquad (x \in D). \tag{19}$$

For successive included x, y, the integer-score gap is at least one, so $\log(p^y) - \log(p^x) \ge T - \eta \cdot x > 0$. Every excluded y has $A \cdot y \ge S+1$ and $\eta \cdot y \ge 0$. Every included product has logarithm less than $T(S+1)$. This preserves the entire ordered prefix and controls its infinite complement. Undo the coordinate permutation. The certificate bound and $(d-1)\varepsilon < 1$ give

$$\log \max_i p_i < T \max_i A_i + 1 < (\alpha + \log 2)(c_d n + 1) + 1.$$

For the lower bound, use the admissible word $W = (0, e_1, 2e_1, \ldots, (n-1)e_1)$. Every realization has $p_j > p_1^{n-1}$ for $j \ge 2$, since e_j is excluded. Thus $h_B(W) \ge (n-1)\log\max(B,2)$. This same set proves the unordered lower bound, and the ordered upper bound applies after choosing any admissible internal order. Since $\alpha = O_d(\log\max(B,2))$ and $n-1 \ge n/2$, the asserted uniform orders follow. □

For $n = 1$, repeated Bertrand intervals give $\log\max_i p_i < \log\max(B,2) + d\log 2$. The prime-interval estimate is an established analytic input, not a new prime-gap result. The bound above does not supply an efficient search for the realizing primes or improve unranking for fixed input bases.

Corollary 10 (A bounded prime budget realizes all prefixes)**.** *For $n \ge 2$ and $B > 0$, restricting the ordered prime bases to*

$$B < p_i, \qquad \log \max_i p_i < (\alpha + \log 2)(c_d n + 1) + 1$$

leaves both the set count $C_d(n)$ and the word count $O_d(n)$ unchanged. Their common asymptotic order is $n^{d-1}(\log n)^{d-1}$. For fixed d, B, primes of $O_{d,B}(n)$ bits suffice for every pattern; some patterns require $\Omega(n)$ bits.

This counts distinct exponent patterns, not realizing prime tuples. For one fixed tuple there is exactly one prefix of each length.

7 Further consequences and limitations

The theorem also determines the average completion multiplicity raised in Wagner's discussion [4, Section 3.5]. Write $\mathcal{C}_{d-1}(\le n)$ for all nonempty corner cuts of size at most n in dimension $d-1$. For each such cut T, let

$$F_n(T) = |\{D : |D| = n,\ D \text{ a corner cut in } \mathbb{Z}^d_{\ge 0},\ D \cap \{x_d = 0\} = T \times \{0\}\}|.$$

Corollary 11 (Average completion multiplicity)**.** *For every fixed $d \ge 2$,*

$$\frac{1}{|\mathcal{C}_{d-1}(\le n)|} \sum_{T \in \mathcal{C}_{d-1}(\le n)} F_n(T) = \Theta_d(\log n).$$

Proof. Restriction to $x_d = 0$ sends every nonempty corner cut to a nonempty corner cut. Its fibres partition the n-point cuts, so the displayed mean equals $C_d(n)/A_{d-1}(n)$. For $d \ge 3$, Theorem 1 and its cumulative consequence give the numerator and denominator orders $n^{d-1}(\log n)^{d-1}$ and $n^{d-1}(\log n)^{d-2}$, respectively. For $d = 2$, there is one one-dimensional cut of each positive size, so $A_1(n) = n$, giving the same conclusion. □

The restriction, fibre identity and lifting argument are established background [4, Lemma 3.5 and Section 3.5]; the corollary records the consequence of the new matching count. It is a mean over all slices, not a lower bound for each slice. For example, $|T| = n$ forces $F_n(T) = 1$.

There is an established geometric interpretation. For a cut D, put $\alpha_D = \sum_{x \in D} x$. For fixed size n, these vectors are distinct and each is a vertex of their convex hull. Indeed, weights strictly cutting off D make $w \cdot \alpha_D$ smaller than $w \cdot \alpha_E$ for every other n-point set E: each replacement point has greater weight than every removed point. Thus α_D is uniquely exposed. Conversely every vertex of the finite hull is one of its generators. The vertex count of the corner-cut polytope is consequently $C_d(n)$. This correspondence is known [1, 5, 9]; in particular, Müller [5, Lemma 2.6] gives the exposed-vertex argument. Our contribution is the quantitative lower count, not the dictionary. For generic experimental designs the same count describes their corner-cut models [9, Definition 7 and Theorem 2].

For fixed B, let $S_{d,B}(n)$ count distinct exponent sets of length n obtained over ordered tuples of distinct primes greater than B. Count each set once, however many tuples realize it. Qualitative realization and the converse observation in Section 1 give

$$S_{d,B}(n) = C_d(n) = \Theta_d(n^{d-1}(\log n)^{d-1}),$$

$$\sum_{j=1}^{n} S_{d,B}(j) = \Theta_d\big(n^d(\log n)^{d-1}\big).$$

The qualitative equality is a consequence of Vietri's earlier theorem. Corollary 10 strengthens it by imposing the explicit, asymptotically optimal worst-case prime-size budget.

The pointwise count gives a matching asymptotic order in every fixed dimension and for the established polytope and model dictionaries. It supplies no asymptotic leading constant, no new enumeration algorithm, no uniform claim as d grows with n, and no fixed-prime unranking complexity improvement. The prescribed-size step is Lemma 4; the earlier cumulative construction alone did not give that conclusion.

8 Related work and reproducibility

The lower bound concerns the same cardinality count as Wagner's (1). The proof uses three ingredients: a uniform finite primitive-frame estimate, injectivity at a prescribed size, and disjoint aspect ranges. We describe the relevant background for these steps. Wagner's upper result also covers faces and flags of the fixed-size corner-cut polyhedron [4, Section 3.5]; we claim no improvement of that upper order. The main advance is the matching lower count. Müller [5, Sections 2–5] studies the same polytopes, including their planar vertex order and adjacency. Her enumeration discussion [5, Section 1, p. 324] cites the planar result and Wagner's bounds; her structural theorems do not supply the bounded-cardinality count proved here.

The unrestricted density of rectangular primitive integer matrices is known [7, 8]. The later paper repairs a defect in the earlier density proof. A centered-cube density alone does not assert the uniform estimate for our moving boxes; Lemma 2 gives a direct finite proof. We claim no new minor criterion or unrestricted density formula.

Chen, Feng, Liu and Wu [17, Theorem 1] give finite probability bounds for appending independent rows to a prescribed primitive matrix. Their random entries use a common interval of length at least the maximum prescribed entry. Our boxes have translated intervals of different lengths, with their shortest length potentially much smaller than the largest entry. That theorem is not a direct substitute for Lemma 2. Prime-by-prime rank analysis and finite determinant cutoffs are established methods; our claimed contribution is their use in

the multiscale corner-cut count. The additional fixed-size step is the two-patch comparison in Lemma 4.

Here is a precise link with finite-grid threshold functions. Put $G_n = \{0, \dots, n\}^d$. An n-point corner cut is exactly an n-set of G_n containing the origin, where an n-set is a subset strictly separable from its complement in G_n. The forward implication follows from the coordinate bound in Section 1. For the converse, write a separator as $w \cdot x < t$, with $t > 0$ since the origin is included. If $w_i \le 0$, all $n+1$ points $0, e_i, \dots, ne_i$ are included, a contradiction. Thus $w_i > 0$. Also ne_i is excluded for every i, so $nw_i > t$; every nonnegative lattice point outside G_n has score greater than t. This proves the converse. Wagner [4, Section 3.2] uses a smaller finite ground set for enumeration; the finite-set reduction itself is not new.

The counting parameter matters even after this identification. Haukkanen and Merikoski [15, Equations (1) and (9)] count all threshold dichotomies of an m-by-r grid, obtaining

$$t(m, r) = \frac{6}{\pi^2}(mr)^2 + O(mr^2) \qquad (2 \le m \le r).$$

This total sums over every possible included cardinality. In particular, $t(n+1, n+1) = \Theta(n^4)$, while already $C_2(n) = \Theta(n \log n)$. A lower estimate for the total does not give a lower estimate for the selected n-point class at the origin. Lemma 4 supplies the prescribed-size control in our construction; the known total-grid count is not being presented as a new result.

Constraining a normal by points on parallel boundary levels also has precedents in digital-plane recognition. Coeurjolly et al. [16, Proposition 6 and Theorems 8 and 11] describe feasible plane parameters using upper and lower leaning polygons, under projection or connected-slice hypotheses. Our argument requires the quantitative conclusion that bounded primitive integer normals are distinguished at every sufficiently large prescribed cardinality. We do not claim the general principle of boundary-point recognition as new.

The Hilbert zonotope refines state polyhedra and supports polynomial-time computation of universal Gröbner bases in fixed dimension [10, Proposition 3.2 and Theorem 3.5]. Its count is an upper bound for a common refinement, not the lower bounded-cardinality count proved here.

Elizalde and Galván discuss higher-dimensional pyramidal partitions and refer to Wagner's growth bounds [19, Section 8.2]. Our comparison uses the preprint version [19]; the journal article is listed separately as [20].

Kalka and Kuhnt [18, Section 2.2, p. 6] explicitly restate Wagner's lower and upper orders for corner cuts, while treating a numerical statistical fan. That passage does not supply a stronger corner-cut counting bound or a proof of our prescribed-size lemma.

Vasiliev and Pavlov [12, Definition 5 and Section 6] call the finite initial intervals of global monomial orders *strictly admissible* orderings. Their parameter is the number of monomials in the interval, and Tables 7–8 enumerate ordered intervals in two and three variables. Forgetting their internal order gives corner-cut sets; their wider finite *admissible* class need not form an initial interval [12, Remark 1]. The finite-weight representation and recursive border/cone enumeration are established background. The full paper supplies algorithms and numerical tables, but no asymptotic corner-cut count, no prescribed-size lower bound of the order in Theorem 1, and no quantitative prime-realization bound. Our contribution concerns the uniform cardinality count, rather than introducing this family of finite orders.

Scope of the comparison. The count in Theorem 1 improves the lower bound of [3, 4] and matches their upper bound. The restriction and lifting viewpoint, primitive-matrix methods, boundary-point recognition, and polytope and model correspondences are attributed above. The improvement is the resulting uniform cardinality count. Theorem 7 extends the count to

complete words in the established finite-ordering class. Theorem 9 bounds prime height rather than asserting a new qualitative realization result. Robbiano's finite-weight consequence, determinant compression methods and Dusart's interval estimate retain their original attribution. The accompanying research audit records precisely which source versions and statements were inspected, including the use of the author's thesis proofs for [3] and the deferred journal-version check for [20]. It is available at `results/research/corner_cut_growth_audit.json`.

Exact checks. The standard-library Python checker `scripts/run_corner_cut_growth_checks.py` uses independent determinants, modular Gaussian elimination and exact rational feasibility. It checks primitive-plane injectivity, the conditional residue bound and a nonprimitive-projection negative control. The prescribed-size checks construct integer Bezout increments and affine-coset reductions, then verify reversed orders on the two patches for 2,280 ordered pairs of distinct normals in dimensions two through six, each at five boundary heights with different residues relative to the witness height. All 11,400 pair–height checks pass. The exact points and scores are saved. The 200 tie-resolution fixtures include 150 partial boundary levels and 50 complete ones. It exhausts stated small boxes in dimensions two through four and uses deterministic samples in dimensions five and six. Six prime-realization fixtures, including equal weights and unused coordinates, are checked by integer products and an independent heap generation of the corresponding smooth-number prefixes. As a check of the finite-ordering identification, all 24 planar entries of [12, Table 7] agree with the exact crossing count. Exhaustive border extension with exact rational feasibility reproduces the eight entries for lengths three through ten in their Table 8. All 1,023 accepted certificates are checked independently by an additive heap; the non-initial finite ordering in their Remark 1 is rejected. Five strict orders of one 50-point planar cut are also checked by a finite-box sort independent of the prefix heap, illustrating the distinction from fixed-vertex maximal face flags. The separate prefix checker tests 230 integer-score certificates and 90 matrix-order prefixes with exact rational vertex enumeration and an independent scalar heap. These include unused coordinates, all active dimensions up to four, pure-axis collision checks and the rational error budget for Theorem 9. They do not experimentally verify the analytic prime-interval theorem. The explicit constants and 352 scale-rounding cases in dimensions two through twelve are checked with rational arithmetic. These computations can falsify steps; they do not prove the uniform theorem or its novelty.

Run, from the repository root,

```
python3 -B scripts/run_corner_cut_growth_checks.py
python3 -B scripts/run_interleaving_research.py --ranks 20000
```

The reports record fixtures, witnesses, exact commands and source hashes:

```
results/research/corner_cut_growth_checks.json
results/research/exponent_interleaving.json
```

The proofs above are independent of the computation. The other project kernels and public rank/unrank interface are unchanged.

Research origins and assistance. This work began with the project originator's smooth-number and interleaving observations and continues under their research guidance. Proof development, literature comparison, verification code and drafting used substantial AI assistance (OpenAI Codex). Full proofs and reproducible checks support independent assessment.

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